Project Proposal

Project Title	Château Collective
Group Name	Château Nexus (P2 - Team 31)
Group Members	Bryan Toh, Ong Bang Xi, Wayne Lee Soo Wei, Leong Yi Phang, Tason Chua Dong Xuan, Cheng Chun Yuan, Oh Jia Rong.

Project Description:
Château Collective is a secure browser-based e-commerce marketplace for first-hand and pre-owned luxury goods such as handbags, watches, jewellery, shoes, wallets and accessories. The application supports guests, buyers, sellers and admins. Buyers can register, log in, browse approved listings, search/filter items, add products to cart, complete a simulated checkout, view their own orders and submit reviews only for completed purchases. Sellers can submit second-hand listings, upload product images, manage their own listings and view sales records. Admins can approve/reject second-hand listings, manage users, handle disputes and review audit logs. Because luxury goods are high-value items, the project focuses on realistic security risks combating counterfeit goods by having in house authentication, seller fraud, account takeover, unauthorized order access, price tampering, fake reviews, file-upload abuse and privilege escalation. Second-hand products remain hidden until admin approval, demonstrating secure workflow control and business-logic protection.

Innovation, creativity & social impact:
Château Collective is designed to apply secure principles to a luxury e-commerce marketplace with workflow controls designed for high-value goods. Unlike a basic buying and selling platform, it includes features such as admin approval for second-hand listings, audit logs, and protection against counterfeit-related abuse. This makes the system more realistic and trustworthy for luxury commerce as counterfeits are rampant. The project tackles real-world challenges faced by luxury marketplaces, such as seller fraud, fake reviews, account takeover, unauthorized access, and price tampering, while presenting them in a practical browser-based application. Its social impact lies in encouraging safer resale of luxury goods by helping reduce counterfeit sales and fraudulent listings, which protects both buyers and sellers. It also promotes responsible digital commerce by showing how secure design can improve trust, transparency, and accountability in online marketplaces.

Technology Stack:

Backend	Python Flask
Frontend	HTML, CSS, Javascript, Bootstrap and Jinja2 templates
Database / ORM / Database Migration	SQLite + SQLAlchemy, Flask-SQLAlchemy + Flask-Migrate, Alembic
Authentication	Werkzeug password hashing and Flask session management
Testing / Automation	Pytest, GitHub Actions, pip-audit
Security Testing	Bandit and Semgrep for SAST, OWASP ZAP for DAST

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Declaration of no usage of existing codes We acknowledge that our secure web app to be developed is not an adaptation of existing codes.

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Cheng Chun Yuan	Tason Chua	Leong Yi Phang	Bryan Toh
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Ong Bang Xi	Wayne Lee	Oh Jia Rong	